

The Japan Factor Puzzle, Part 2: The Rescue That Wasn't Quite

Code and data: [<https://github.com/GT-HLab/quant-research/tree/master>].

Over the same thirty-five years that turned one dollar of Japanese momentum into \$1.10, one dollar of Japanese value became about \$4.50.

Part 1 left momentum flat on its back in Japan, a t-statistic of 0.59, indistinguishable from zero across three decades and three global regimes and refused every easy explanation: not a weak index, not one bad decade, not statistical noise disconnected from the global factor. That question may not have a clean data answer. But there is a more useful one to ask. A broken factor in isolation is not the same as a broken factor in a portfolio, and nobody holds momentum alone. It usually rides alongside value, and the two are natural opposites: momentum needs trends to persist, value needs prices to revert.

That raises a possibility worth checking. If momentum and value tend to struggle at different times, a market that punishes one might reward the other, and a portfolio holding both could hold up even where a single leg collapses. It is at least plausible that Japan's momentum problem dissolves once you stop looking at momentum on its own.

If Japan is the market where momentum breaks, theory says it should be the market where value works.

The Japanese value premium is real

All results use the Kenneth French developed-markets library: monthly value (HML, high-minus-low book-to-market) and momentum (WML, winners-minus-losers) factor returns for North America, Europe, Japan, and Asia Pacific ex Japan, November 1990 through mid-2026. Returns are raw long-short factor returns; the portfolios are self-financing, so no risk-free adjustment applies. No transaction costs, no implementation constraints. The same late-1990 start caveat from Part 1 carries over.

		Ann. Mean (%)	Ann. Vol (%)	Sharpe	t-stat	Max DD (%)	Max DD Date
Region	Factor						
North America	Value	1.72	12.06	0.14	0.85	-60.51	2020-09-30
	Momentum	6.63	15.71	0.42	2.52	-49.97	2009-09-30
Europe	Value	4.25	9.02	0.47	2.81	-50.89	2020-10-31
	Momentum	10.58	13.13	0.81	4.80	-45.59	2009-09-30
Asia ex Japan	Value	7.23	10.46	0.69	4.12	-28.09	2020-12-31
	Momentum	10.48	14.31	0.73	4.37	-52.81	2000-11-30
Japan	Value	4.85	10.77	0.45	2.68	-46.46	2020-11-30
	Momentum	1.43	14.48	0.10	0.59	-41.96	2012-02-29

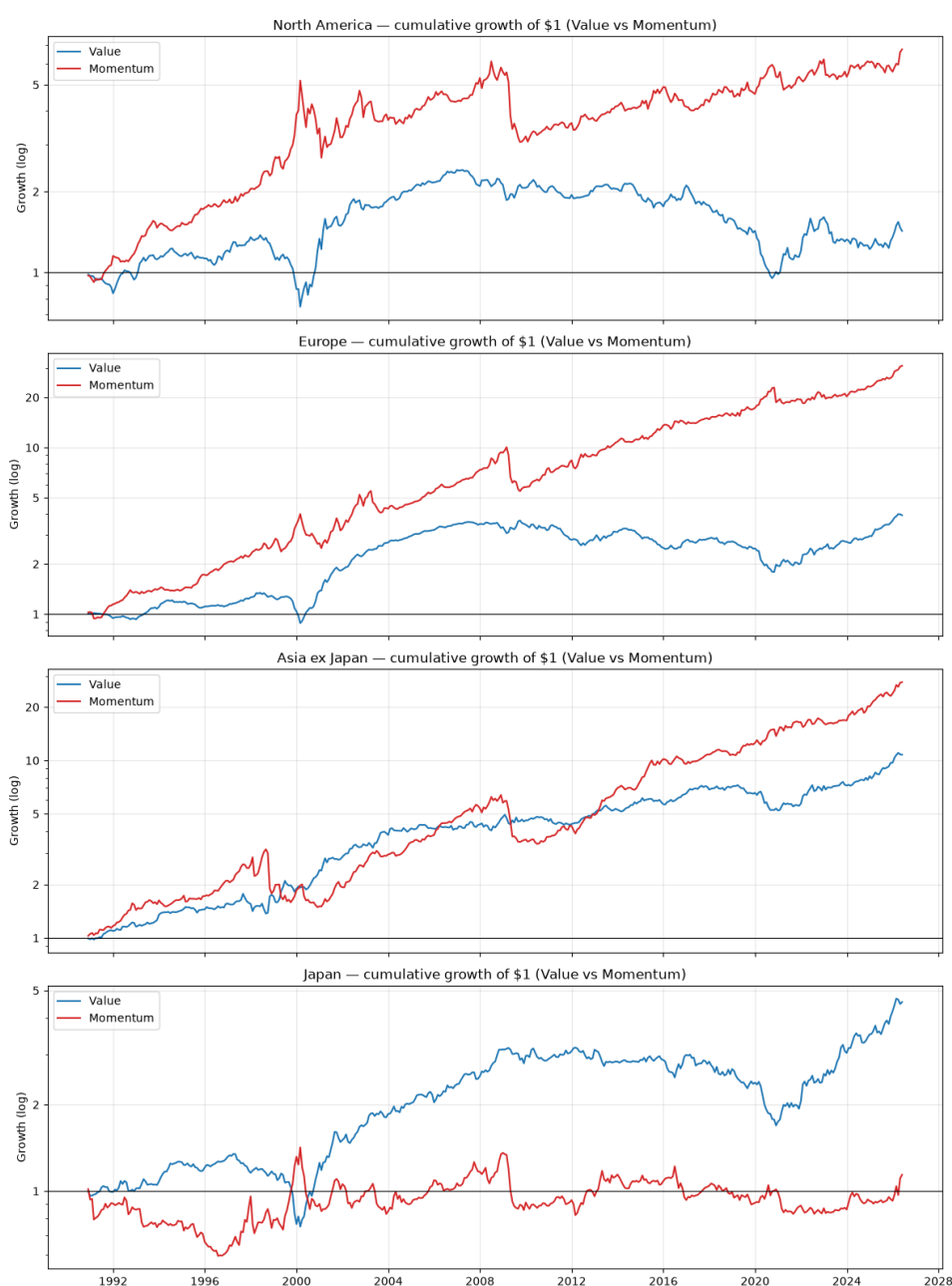
[Table 1: Value factor summary statistics per region]

Japanese value delivered 4.9% annualized at a 0.45 Sharpe, with a t-statistic of 2.68 significant by the classical threshold, and a world apart from Japanese momentum's 1.4% and $t = 0.59$. On this measure Part 1's setup holds: value works in Japan where momentum does not.

But two things complicate the tidy picture.

First, Japanese value is not exceptional globally. Asia ex Japan posts a stronger value premium (7.2%, $t = 4.12$), and Europe's is comparable (4.3%, $t = 2.81$). Japan is a solid value market, not a standout one and that matters. The standard rescue argument needs Japan to have an *unusually* strong value premium, one lopsided enough to carry the missing momentum leg. It does not. It has an ordinary one, smaller than Asia's, barely larger than Europe's. Whatever rescues Japan, it will not be an exceptional value leg, because Japan doesn't have one.

Second, value's premium comes with the deepest kind of pain: a maximum drawdown of -46% , bottoming in November 2020, the COVID growth melt-up that crushed value everywhere. In fact all four regional value factors hit their worst losses within three months of each other in late 2020 (Table 1). Where momentum's crashes are idiosyncratic to the cycle, value's are global and simultaneous. That distinction matters for anyone hoping to diversify across regions rather than across factors: momentum lets you, value does not.



[Graph 1: Growth of \$1 — Value vs Momentum, by region, log scale]

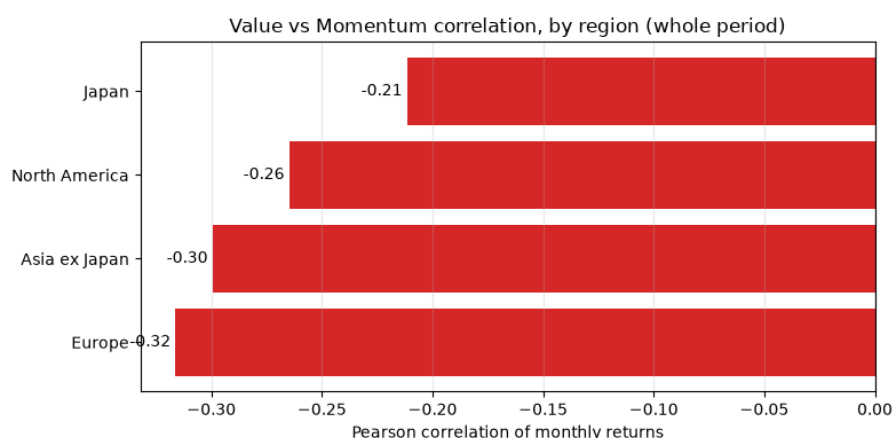
Compounded, the regional contrast is stark. In North America, Europe, and Asia ex Japan, momentum (red) buries value (blue). Japan is the lone inversion: value climbs to roughly \$4.50 while momentum flatlines near \$1. The picture that made momentum look broken in Part 1 makes value look like the answer. The rest of this piece is about why that answer is only partial.

The offset mechanism

So value alone is not the rescue, and momentum contributes almost nothing. That leaves only the interaction between them, the hope that even two unexceptional legs, combined, add up to something worth holding. This is where the rescue actually lives: not in either factor, but in how they move against each other.

What matters now is the correlation between the two, whether the two legs tend to move together or apart, and the logic is straightforward. If value and momentum tend to fall in the same months, pairing them buys nothing. If they tend to move apart, one strong when the other is weak, the combination is smoother than either alone, and the more consistently they diverge, the smoother it gets. That tendency to move apart is what turns two ordinary factors into a portfolio worth more than its parts.

So the whole case for Japan now rests on that correlation. And Japan, of the four regions, is the one that needs it most: it is the only market asking the pairing to carry a leg that pays nothing, rather than merely to smooth two that already pay. If the offset is real anywhere, it has to be real here.



[Table 2: Value–Momentum correlation, by region]

Value and momentum returns are negatively correlated in every region, in a narrow band from -0.32 in Europe to -0.21 in Japan.

Japan sits at the wrong end. The region with the most riding on the offset has the weakest one of the four. The market that most needs value and momentum to pull apart is the market where they pull apart the least.

The 50/50 combo: a risk story, not a return story

The test that matters is the portfolio. For each region we build an equal-weight, monthly-rebalanced 50/50 value-and-momentum combination.

	Value Sharpe	Momentum Sharpe	Combo Sharpe	Combo Ann. Mean (%)	Combo Ann. Vol (%)	Combo t-stat	Combo Max DD (%)	Combo Max DD Date
Region								
North America	0.14	0.42	0.49	4.18	8.54	2.92	-31.76	2020-12-31
Europe	0.47	0.81	1.11	7.41	6.69	6.61	-19.20	2010-02-28
Asia ex Japan	0.69	0.73	1.18	8.85	7.49	7.05	-26.35	2009-08-31
Japan	0.45	0.10	0.39	3.14	8.06	2.32	-34.49	2021-08-31

[Table 3: Sharpe standalone legs vs 50/50 combo + summary statistic]

In North America, Europe, and Asia ex Japan, the combo Sharpe ratio beats **both** standalone legs. Europe rises from 0.81 (momentum) to 1.11, Asia from 0.73 to 1.18. This is textbook diversification, and it is exactly what the rescue argument predicts: pair a working leg with a struggling one and the combination outperforms either alone.

Japan behaves differently, and the difference is categorical rather than marginal.

On the narrow claim, the rescue works: Japan's combo Sharpe of 0.39 is a real, live number that crushes standalone momentum's dead 0.10. A value-and-momentum portfolio is investable in Japan even though momentum alone is not.

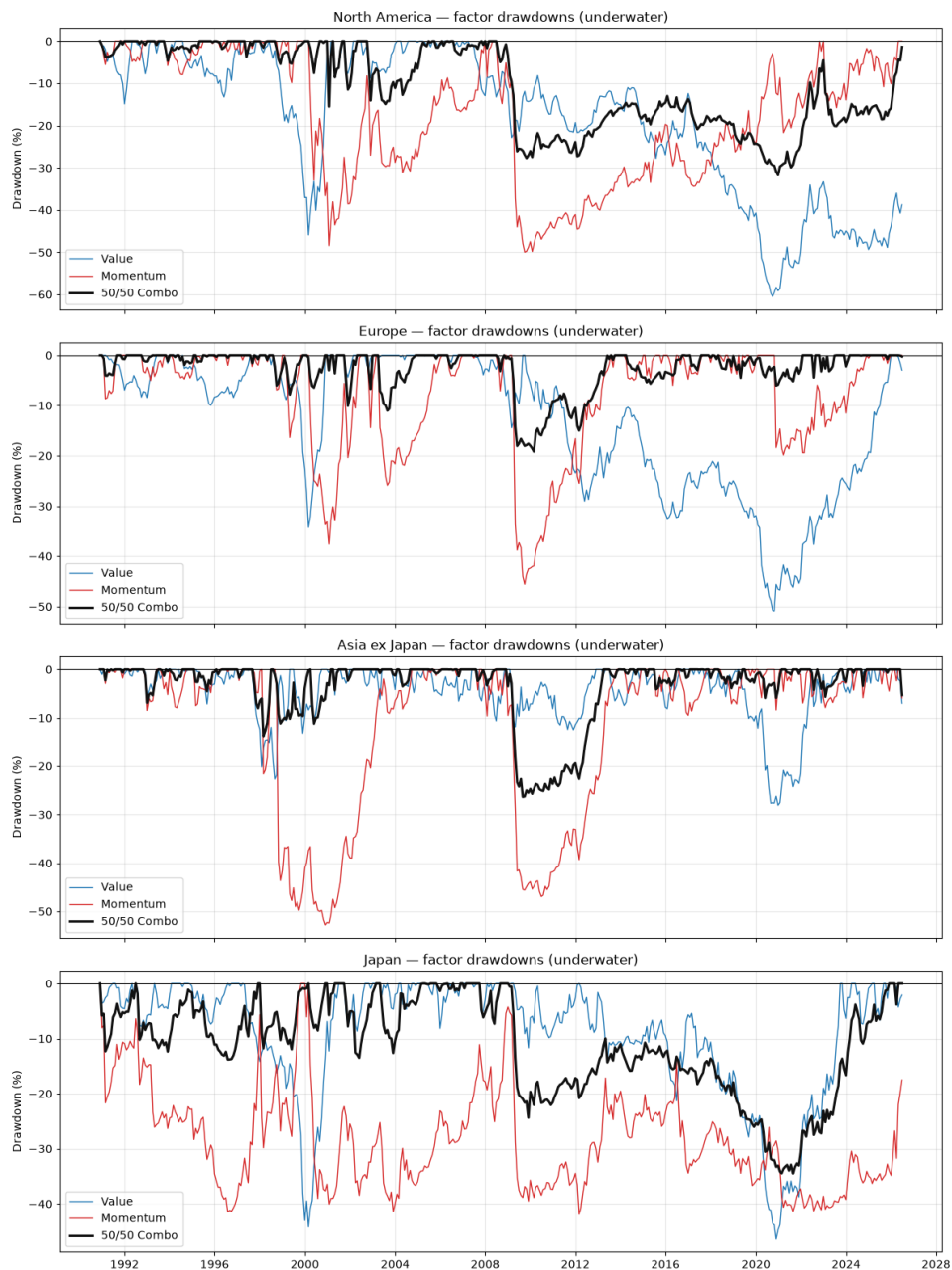
But Japan is the only region of the four where combining the factors *lowers* risk-adjusted return. Elsewhere the combination adds 0.07, 0.30, and 0.45 of Sharpe over the better standalone leg. In Japan it subtracts 0.06. The combo's 0.39 trails standalone value's 0.45. It is also the lowest combo Sharpe of the four regions.

The combination halves value's premium by averaging in a leg that returns 1.4% a year: Japan's combined mean falls to 3.1% from value's 4.9%. The volatility does drop to 8.06% from 10.77% but at a correlation of -0.21 it does not drop enough to pay for the return surrendered. The weakest offset in the sample, applied to the most lopsided pair of legs in the sample, dilutes faster than it smooths. In Japan the dilution wins.

Where the combo earns its place in Japan is risk.

	Value Max DD (%)	Momentum Max DD (%)	Combo Max DD (%)	Combo vs Deeper Leg (pp)
Region				
North America	-60.50	-50.00	-31.80	28.80
Europe	-50.90	-45.60	-19.20	31.70
Asia ex Japan	-28.10	-52.80	-26.40	26.50
Japan	-46.50	-42.00	-34.50	12.00

[Table 4: Maximum drawdown, Value / Momentum / Combo]



[Graph 5: Factor drawdowns (underwater), by region]

In every region, the 50/50 combo's worst drawdown is shallower than either leg alone. The effect is large: North America's worst loss falls from -60% (value) and -50% (momentum) to -32%; Europe's from -51% and -46% to -19%. Japan is no exception its combo drawdown of -35% is meaningfully shallower than value's -46% or momentum's -42%.

So the offset in Japan is real but small: large enough to clip twelve points off the worst tail, too small to justify the return dilution. Whether that trade is worth making is not a question the data settles. Both readings are defensible.

Even here, though, Japan places last. Europe's combination cuts 31 points off its worst drawdown; Japan's cuts twelve. On the one dimension where pairing value and momentum works in Japan, it still works less well than anywhere else.

Resolve, and deepen

Two things are settled and one is not.

Resolved. Value pays a significant premium in Japan; value and momentum are negatively correlated there; and a combined portfolio is investable where standalone momentum is not. The global factor framework survives Japan. You would not have abandoned factor investing in Japanese equities, you would have leaned on value. The instinct behind the rescue argument: look at the pair, not the leg, is a better way of framing the problem.

Qualified. In Japan the pair did worse than the better leg alone on a risk-adjusted basis, and the honest case for holding both is a drawdown case, not a return case. The rescue is real, but it is a rescue of the ride rather than the destination.

Deepened. The offset explains why a portfolio holds up. It does nothing to explain why the standalone momentum premium vanished in the first place. That puzzle from Part 1 is untouched: a factor carrying the full crash risk of the global momentum trade while paying nothing is still unexplained, and the value combination reframes the practitioner's problem without closing the theoretical one.

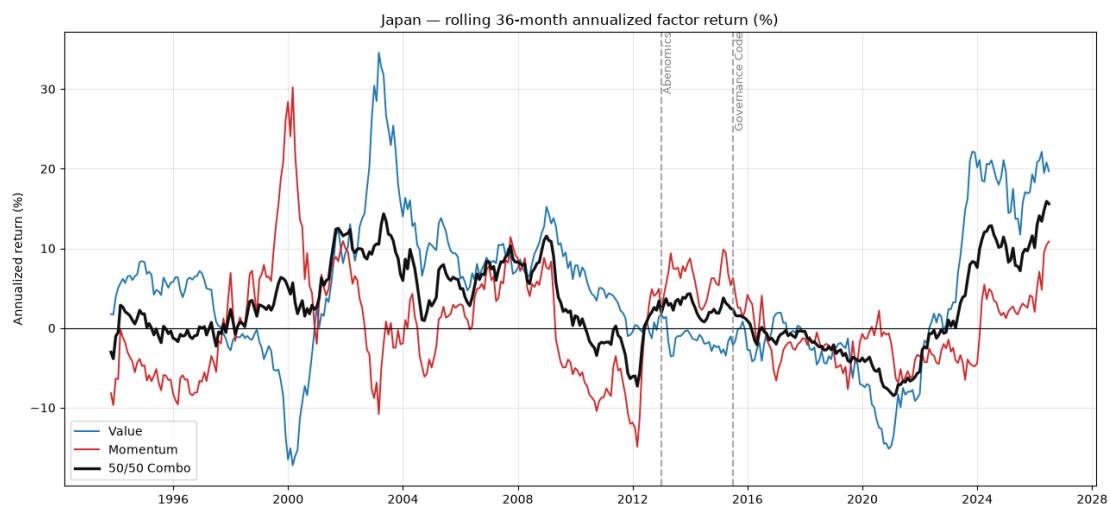
There is also a hint worth flagging, though this data cannot confirm it. Japan has both the weakest momentum premium and the weakest value–momentum offset. Those two facts may share a cause, whatever suppresses trend persistence in Japanese equities might also weaken the relationship between trend and valuation. That is a hypothesis, not a finding, and it is the sort of thing that requires a different test than the ones run here.

A coda on reform: the tailwind that wasn't

One loose end sits apart from the combination question and is worth recording before closing. It concerns the value leg on its own, not how it pairs with momentum, but whether it has held up over time. It hasn't, and it faded through precisely the years designed to strengthen it.

Part 1 promised to ask whether the governance-reform era changed the Japanese value premium. To see why anyone expected it to, we have to understand what the reforms were fixing. For two decades after the 1990 crash, Japanese equities carried a structural pathology that was poison for value investing specifically. Thousands of companies traded below the book value of their own assets and stayed there indefinitely, because nothing forced the discount to close. Three frictions kept it that way: corporates hoarded cash instead of returning it, leaving return on equity chronically low; cross-shareholdings, banks, suppliers, and customers holding permanent blocks of each other's stock, left much of the share register inert and loyal to management; and boards, with few independent directors, faced no consequence for underperformance. Value depends on a catalyst: buy the discount, wait for it to close. In Japan there was none. Cheap stocks were simply cheap forever.

Starting in **2012**, a wave of reform set out to install the missing catalyst. Abenomics made shareholder returns a government priority; the Stewardship Code (**2014**) pushed institutions to actually engage and vote their shares; and the Corporate Governance Code (**2015**) hit the frictions directly, pressing companies to add independent directors, unwind cross-shareholdings, and answer for their return on equity. Every one of those mechanisms should reward a strategy that buys cheap book value and waits for it to be realized. A value investor in 2015 would have made a clean prediction: the Japanese value premium is about to strengthen.



[Graph 6: Japan rolling 36-month annualized factor return]

Region (Value factor)	1990-1999		2000-2012		2013-2026	
	Ann. Mean (%)	t-stat	Ann. Mean (%)	t-stat	Ann. Mean (%)	t-stat
North America	-1.11	-0.35	7.58	2.02	-1.77	-0.54
Europe	0.60	0.30	8.27	3.06	2.73	1.02
Asia ex Japan	7.78	1.83	8.01	2.96	5.72	2.25
Japan	-2.46	-0.82	11.37	3.87	3.55	1.12

[Table 4: Value premium by sub-period]

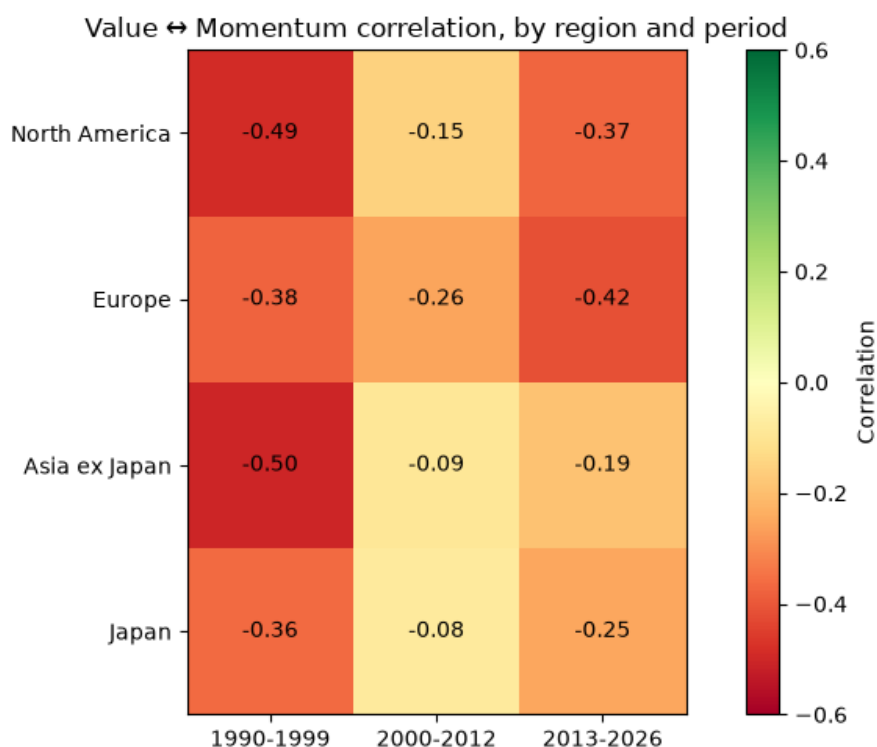
The premium moved in the wrong direction. Japanese value earned 11.4% annualized in the **pre-reform** 2000–2012 window ($t = 3.87$), its strongest era on record. Through the reform years it faded to 3.5% and a t-statistic of 1.10, no longer distinguishable from zero. The rolling series shows value's best stretch ending well before the reforms and turning sharply negative through 2013–2021, recovering only after 2022. The reforms that should have been value's tailwind coincided with its worst run in twenty years.

The post-2022 recovery is its own complication, and points to a fuller story rather than a simpler one. In early 2023 the Tokyo Stock Exchange launched a far blunter initiative than anything in the earlier codes: it publicly pressured companies trading below book value to disclose concrete plans to fix it, and began naming those that complied. Japanese value and equities broadly rallied hard from that point, and the timing lines up with the recovery in the rolling series. So the honest shape of the evidence may be that the 2012–2015 reforms underdelivered while a later, more coercive push finally bit. Which, if anything, sharpens the puzzle: the softer governance codes, the ones that best fit the tidy story of a strengthening value premium, are the ones that failed to move it.

Two reasons not to over-read this. Value fell nearly everywhere in this window, North American value returned -2.0% annualized over 2013–2026 and European value 2.85% , so Japan's decline is real but not obviously Japan-specific, and separating a domestic policy effect from a global growth-stock regime is beyond what these series can do. And sub-period t-statistics are built on short samples; a decade cannot cleanly separate a fading premium from an unlucky one, the same statistical humility Part 1 insisted on for momentum. The most that can be said is narrower and still uncomfortable: if governance reform strengthened the Japanese value premium, no trace of that strengthening appears in the factor returns.

Where this goes next

The offset is not just weak in Japan, it is unstable everywhere. Split the sample by regime and the negative correlation nearly vanishes in the middle window: across all four regions it compresses toward zero in 2000–2012, falling to -0.08 in Japan and -0.09 in Asia ex Japan, precisely the crisis decade when an investor most needs value and momentum to diverge. The weakening is global, not Japanese, and it is exactly the wrong behavior at the wrong time, the hedge thins out just as the drawdowns it is meant to cushion arrive.



[Graph 5: Value–Momentum correlation, by region and sub-period]

A fixed 50/50 can do nothing about this. It holds the same weights whether the offset is -0.50 or -0.08 , and whether momentum is paying or dead. If the two factors take turns working, and their relationship shifts with the regime, a static split leaves return on the table and holds a hedge that is not there when it counts. Part 3 asks whether *timing* the exposure, leaning toward whichever factor the regime favors, rather than freezing the split at half and half, rescues in Japan what equal weighting could not, or whether the same broken momentum leg defeats the timed portfolio just as it defeated the fixed one.

Limitations

Sample begins November 1990. Long-short academic factors ignore shorting costs, borrow availability, and financing; decile sorts are not investable portfolios. Momentum turns over faster than value, so the net-of-cost case for the momentum leg is weaker than these gross numbers suggest, not stronger.

Value is one definition (HML, book-to-market) and momentum one (WML, 12-1), on four regional aggregates. The 50/50 weight is deliberately naive.

Correlations between independently sorted French factors understate the offset available from a matched-factor construction; the sign and the cross-regional ranking, not the precise coefficient, are what to trust.

And the central caveat from Part 1 holds on both legs now: a high t-statistic supports a premium, a low one is a failure to reject zero, not a demonstration of it.

Personal Research : views are my own